



A critique of ‘Strong Beginnings’ initial teacher education reforms: mandating neuroscience as core curriculum within the ‘what works’ movement

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Received: 13 November 2023 / Accepted: 1 June 2024
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Abstract

The paper studies the rise of neuroscience in initial teacher education, paying attention to the relatively recent Australian Government (2023) report titled ‘Strong Beginnings’. In taking up a critical policy sociology lens, we focus on the first priority within the reforms, which is mandating brain science and the ‘brain and learning’ as core curriculum within initial teacher education. The reforms will embed standardised curriculum into initial teacher education and tie this curriculum to graduate teacher standards, bracketed within prescribed texts, ideologies and agendas. The reforms are positioned within the ‘what works’ movement, increasing accreditation and certain types of evidence, and the role and authority of intermediary organisations, including the Australian Institute for Teaching and School Leadership (AITSL). It promotes research that has been paid for by the think tank Centre for Independent Studies and aligned with the Australian Education Research Organisation and Education Endowment Foundation. In this paper, we endeavour to highlight how the singular and narrow focus on brain-based approaches is not only reductionist, but also potentially generative of oppressive technologies. The mandating of standardised curriculum and brain science undermines educators, including initial teacher educators, and bolsters private interests in education. The standardisation of core curriculum, which will be tied to accreditation processes and graduate teacher standards, is underpinned by a punitive-accountability based approach. Furthermore, whilst it is less visible, these reforms contain brain science tropes redolent of eugenics and deficit framings of low socio-economic status students.

Keywords Neuroscience · Brain science · Initial teacher education · What works · Evidence-based reform · Policy

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Published online: 09 July 2024

Springer

Introduction: initial teacher education reforms

In Australia, and globally, there have been ongoing major reviews into Initial Teacher Education (ITE) (Alexander & Bourke, 2021; Brennan & Willis, 2008; Cochran-Smith, 2005; Mayer, 2014; Mockler, 2023) with ‘101 government inquiries into Australian teacher education since 1979’ (Louden, 2008, p. 357). For this reason, Mockler (2023) describes Australia as ‘very fertile terrain for examining ITE policy reform’ (p. 115). The reforms of initial teacher education occur in the context of global competitiveness, marketisation of school systems and the rise of global ‘evidence brokers’ such as the OECD (Lewis, 2017), in addition to federalism, as national governments have sought increasing control over their education systems to ensure a greater level of alignment with policy priorities (Grek & Ozga, 2010; Lingard, 2000; Savage, 2016).

In Australia, these significant reviews as initiated by the national government and led by selected panel members, have focused on carrying out major reforms of the initial teacher education sector, as we have pointed to earlier (Rowe and Skourdoumbis, 2019) claiming there is a need for ‘urgent national action to improve the quality of initial teacher education’, in order to improve ‘teacher quality’ and ‘classroom readiness’ (TEMAG, 2015). These reforms have resulted in a number of major changes including mandating new accreditation processes for initial teacher education providers, introducing new forms of assessment, namely the teaching performance assessment (TPA) and the introduction of a ‘user-pays’ literacy and numeracy test for preservice teachers (LANTITE), described by Barnes and Cross (2021) as a ‘gate-keeping mechanism’ (p. 455). Overall, the interventionist reviews into ITE have retained ‘notable impact’, including impacts on professional standards, certification, professional development, assessment and practicum design (Rowan et al., 2015, p. 274).

In this paper we focus our attention on the most recent major reforms, at time of writing, ‘Strong Beginnings: Report of the Teacher Education Expert Panel’ (Australian Government, 2023), a panel which was appointed in September 2022, in response to government roundtables identifying a critical teacher shortage. ‘Strong Beginnings’ and the associated reforms continue similar logics of the 2015 ‘Action Now’, and 2022 ‘Quality Initial Teacher Education’ reviews as summarised below (see Table 1), yet maintains an explicit difference via its focus on neuroscience and brain science, which we elaborate upon subsequently (see Table 1):

In the following section, we briefly discuss the four ‘priority reforms’ set out by the panel (Australian Government, 2023). In this paper, our analysis focuses on the first priority, which is mandated standardised curriculum, to be ‘embedded into all ITE programs by late 2025’ (Mockler, 2023, p. 128) and its emphasis on neuroscience, or ‘the brain and learning’ as core content.

Four ‘Priority reforms’: why the shift towards neuroscience is notable

The ‘Strong Beginnings’ (2023) report puts forth four priority reforms, with each reform carrying recommendations (14 in total) (Australian Government, 2023,

Table 1 This table briefly summarises major Australian Government reports into Initial Teacher Education since 2015

Key National Reports focusing on Initial Teacher Education Reform	Brief summary of recommendations/reforms
2015 Teacher Education Ministerial Advisory Group (TEMAG), titled: “ <i>Action Now: Classroom Ready Teachers</i> ” (TEMAG, 2015)	The introduction of major reforms including: LANTITE (Literacy and Numeracy test for pre-service teachers); Teaching Performance Assessment (TPA); Mandating new accreditation processes
2022 QITE Review: Quality Initial Teacher Education Review, titled: “ <i>Next Steps: Report of the Quality Initial Teacher Education Review</i> ” (Australian Government, 2023)	Seventeen recommendations in total (see page iii, iv, v). This includes raising the status of teachers; funding for ITE should be based on performance; and the TPA should be expanded
2023 “ <i>Strong Beginnings: Report of the Teacher Education Expert Panel</i> ” (Australian Government, 2023)	Fourteen recommendations in total (see page 17, 18, 19). These will be analysed in further depth in this paper

Table 2 This table summarises the four reform areas (‘Priority Reforms’) (see, Australian Government, 2023)

Priority Reform	Summary
Priority Reform 1:	Strengthening ITE Programs to deliver effective beginning teachers – establish the core content and mandate it in national accreditation The Panel has defined four types of core content: 1. The brain and learning 2. Effective pedagogical practices 3. Classroom management 4. Responsive teaching
Priority Reform 2:	Strengthening the link between performance and funding for ITE programs
Priority Reform 3:	Improving the quality of practical experience in teaching
Priority Reform 4:	Improving postgraduate initial teacher education programs to attract more mid-career entrants

pp. 17–21). We utilise a table to summarise the four priority reforms below (see Table 2):

We focus on priority reform number one, which is to establish ‘core content for ITE programs’, as mandated content in initial teacher education courses (Australian Government, 2023, pp. 9, 10, 95, 96). The core content is specified as four different types, including the brain and learning, effective pedagogical practices, classroom management and responsive teaching (p. 17). ITE providers will need to design courses that ‘coherently incorporate the core content’ and provide evidence of this, during their accreditation process (p. 17). The core content will be embedded and tied to the graduate teacher standards (1.1 and 1.2) and graduates will need to ensure that they are competent within the core content (p. 98). The Australian Institute for Teaching and School Leadership (AITSL) will take a leading role in this regard, and from reading the review and its recommendations, their scope of powers, role

and assessments will be bolstered under these reforms. Overall, the report maintains a strong emphasis on *evidence* or evidence-based (used 200 times in the 120-page report); *accreditation* (used 197 times); and *assess/assessment* (used 90 times). For these reasons, and as we elaborate upon further, the logics underpinning the reforms are located within the ‘what works’ movement (Abdo et al., 2021).

In this paper we focus on the ‘brain and learning’ as core content (Australian Government, 2023, pp. 9, 10). Significantly, we draw upon the explicit detail that is provided in relation to the core content within the Appendix of ‘Strong Beginnings’ (see Appendix D: Core Content for ITE Programs, ‘Core Content 1- The Brain and Learning’, p. 95, 96). The report itself does not explicitly draw upon the term ‘neuroscience’, and nor does it define this utilisation, but the recommended texts in the core content indicates a focus on neuroscience, neuromyths, and educational psychology (p. 96). We take up the term ‘neuroscience’ in the following analysis, referencing the OECD (2007) report¹ to indicate how neuroscience is shaped by multiple disciplines, encompassing ‘neurobiology, cognitive neuroscience, behavioural neuroscience, cognitive psychology etc.’ (OECD, 2007, p. 21).

The reasoning behind this focus on neuroscience is multiple; first of all, we argue that the emphasis on neuroscience within initial teacher education, and the level of authority and remit this is attached to, is significant and important. Whilst the neuroscience reasoning is far from new or novel, being present within early childhood reforms since the 1960s in Australia (Millei & Joronen, 2016), this is the first time it has been mandated within initial teacher education nation-wide, and thus represents an important turning point. Second of all, and in relation to the first, the ‘Strong Beginnings’ reforms indicate growing influence of the ‘what works’ movement, demonstrating increasing impact by intermediary organisations (IOs), evidence brokers and conservative think tanks, which we will argue is critically important to illuminate. Indeed, the reforms we focus on were principally influenced by intermediary organisation, the Australian Education Research Organisation (AERO).

The aim of this paper is to critique the ‘brain-based’ approach, grounded within an analysis of the ‘Strong Beginnings’ reforms (Australian Government, 2023); in doing so, we question neuroscience in education as a ‘value-free’ approach, drawing on critiques and limitations of brain-based education; we put forward the risks of a brain-based approach as potentially dehumanising for students, establishing oppressive technologies as located within eugenics; so too, this approach discounts other forms of teacher knowledge and teacher expertise, including teacher educators. As Alexander and Bourke (2021) write, ITE reforms have historically resulted in ‘increased marginalisation of teacher educators’ (p. 471) and a ‘deleterious impact on professional status’ (Larsen & Mockler, 2023, p. 1063). Barnes (2021) points to how teacher candidates believe the ongoing reforms reinforce ‘a deficient view of teachers’ (p. 8).

In the second part of our critique, we examine the policy reforms in relation to the intermediary organisations and evidence-brokers that have directly authored and influenced them (AERO), and in relation to the ‘what works’ movement in which it

¹ ‘Strong Beginnings’ cites the earlier OECD (2002) report ‘Understanding the Brain’ but does not engage with or cite the later OECD (2007) report titled ‘Understanding the Brain’.

is positioned. We point out how these reforms will directly bolster the role, authority and remit of intermediary organisations (IOs), evidence brokers and private–public partnerships, including AERO, but also conservative think tank (Centre for Independent Studies), and the Australian Institute for Teaching and School Leadership (AITSL).

In the following section, we briefly discuss the context in which the report was initiated; setting out that whilst the report was initiated in response to a critical teacher shortage, it fails to respond to it.

Critical teacher shortage

Importantly, the 'Strong Beginnings' review was initiated in the context of a critical teacher shortage (Australian Government, 2023), in that a lack of teacher supply is currently facing Australian schools nation-wide (AEU & IEU, 2022), an issue shared by many OECD countries such as England and the United States (Heffernan et al., 2022). However, whilst the review was initiated in response to a critical teacher shortage, it does not directly engage with this problem and makes very little reference to teacher supply or reasons for high teacher attrition, such as excessive workload, as identified within AITSL surveys (AITSL, 2021) and peer-reviewed research (Heffernan et al., 2022).

Notably, the preceding 'Quality Initial Teacher Education' review (2021) addresses workload, specifying the reduction of 'workload' as a key recommendation (number 3, p. vi) and that 'red tape should be removed' from teachers' work (p. vi).² However, in contrast, 'Strong Beginnings' does not make any recommendations in regard to decreasing workload, and only makes two minimal references to heavy workload in the body of the report (p. 64 and p. 66), which are in relation to improving the practicum experience of pre-service teachers.

In sidestepping this issue altogether, of an excessive workload for teachers which is reportedly driving teacher attrition (Creagh et al., 2023; Fitzgerald et al., 2018; Heffernan et al., 2022; Stacey et al., 2020, 2023), the reform agenda focuses on standardised core content, coupled with improving in-school supervision practicum experiences, as based on a set of nationally agreed guidelines yet to be determined.

Theoretical framework and methodological approach: what's the 'problem' represented to be?

In this paper, we take up Bacchi's (2009) 'what's the problem represented to be?' (WPR) approach to study and critique policy. Whilst Bacchi (2009) may not directly align her approach with critical policy sociology, we argue that the analysis is theoretically related, in questioning power, knowledge production and policy mobilities (Ozga, 2021). In applying and interpreting Bacchi's approach,

² However, the recommendations in their entirety do not reflect this aspiration to remove red tape, for many of the recommendations are focused on *increasing* accreditation standards, punitive approaches and standardised assessments.

we take up questions to guide the critique and analysis (see, Bourke et al., 2020; Mockler, 2023):

1. What's the 'problem' represented to be in a specific policy?
2. What presuppositions or assumptions underlie this representation of the 'problem'?
3. How has this representation of the 'problem' come about?
4. Where are the silences?
5. What effects are produced by this representation of the 'problem'?
6. How/where has this representation of the 'problem' been produced, disseminated and defended? How could it be questioned, disrupted and replaced? (Bacchi, 2009, p. 2)

We pay attention to how problems are framed and constructed (Bacchi, 2009), seeking to interrogate how policy reforms, such as 'Strong Beginnings' represents and socially constructs certain assumptions or presuppositions. Herein, the 'problem' takes on authority and takes an active role in 'constructing' the problem (O'Connor & Rudolph, 2023, p. 9).

The 'Strong Beginnings' reforms rest on *assumptions* of declining teacher quality and classroom readiness, and in the process silences teacher voice, omitting issues of excessive teacher workload. We want to emphasise that the problems that these reforms construct, including declining teacher quality and classroom readiness, do not reflect peer-reviewed research (Gore et al., 2016; Mayer et al., 2012, 2013, 2015), and nor are they a reflection of Graduate Outcomes Surveys (GOS, 2023). Overall, despite advocating for 'evidence-based', the report demonstrates quite weak understandings of quantitative 'evidence'; for example, when making the argument that beginning teachers are not well prepared, 'Strong Beginnings' (2023) rests on the 2022 Graduate Outcomes Survey, citing that *32 per cent* of responses reported a need for 'better and more relevant course content'; *31 per cent* of beginning teachers reported they would have benefited 'from a greater focus on more practical experience'; and *25 per cent* of responses indicated they would have 'benefited from more practical knowledge' (Australian Government, 2023, p. 23). When considering 'evidence' from positivist, traditional or quantitative constructs, as advocated in this report, this would reflect a weak argument and more so an ideological agenda. The actual results of the 2022 Graduate Outcomes Survey shows high levels of 'overall satisfaction by course level and study area' for 'teacher education', with a ranking of 75% overall satisfaction, which is comparatively high when comparing to a range of courses (GOS, 2023, p. 30). Teacher education also ranks well in regard to graduate employment outcomes (p. 11), and median starting salaries (GOS, 2023, p. 12).

Taking up this critical starting point to our analysis, we proceed as follows: we identify a central assumption that the brain-based approach relies upon, which is a 'value-free' science, which justifies deficit framings of students, potentially establishing oppressive technologies; second, we analyse policy mobilities, politics and power in considering how the solution to the problem has acquired

validity and remit, exploring the invested interests that elevate particular forms of knowledge over others. We consider how the brain-based approach was influenced and authored by intermediary organisation (AERO) and has further solidified and strengthened the role and powers of AITSL, pointing to demonstrable influence by conservative think tank CIS.

Neuroscience and brain-based approaches: a value free science?

In asking, what presuppositions or assumptions underlie this representation of the 'problem' (Bacchi, 2009), the narrow emphasis on neuroscience and brain-based approaches to learning rests on numerous assumptions, but one that is quite clear is the ideology that a brain-based approach is value-free, and a 'silver bullet' solution (Lewis & Hogan, 2019). These reforms represent a struggle over knowledge, undermining teacher expertise and teacher knowledge, whilst elevating supposedly 'neutral' and 'value-free' brain-based approaches to learning.

Within this priority reform area (the brain and learning), it specifies three foci: 'novice versus expert learners'; second, 'how the brain learns and retains information'; and third, 'how the brain masters knowledge' (p. 95). It stipulates that ITE students should learn explicit distinctions between the 'brain of a novice vs the brain of an expert' (p. 95), and this binary should be taught to ITE students 'with reference to... mental models and schemas' (p. 95). ITE students should learn about the 'features of biologically primary vs biologically secondary knowledge acquisition' (p. 95). There are numerous references throughout that are explicitly opposed to self-directed approaches, and inquiry-based learning (including the recommended textbooks, p. 96).

The core content that is to be included in teacher education courses is reductionist in its scope in that it explicitly renders learning in narrow frames, as a brain-based activity rather than as a multi-faceted process that incorporates several interactions, including relationships (such as Vygotsky's sociocultural theory that argues learning is a social process, and social interaction forms a key role in learning, and learning is influenced by context). Clearly, learning involves cognitive functions and brain processes; but the strict emphasis on brain science 'cannot be invoked to explain away sociology; instead there will be a dialectical interaction between them' (Rose & Rose, 1976, p. 121). The OECD (2007) supports this argument, writing that 'we by no means claim that neuroscience is a panacea' and rather a 'genuine trans-disciplinary approach drawing from many disciplines is required', including 'education itself or in other social science reference disciplines or indeed in philosophy' (p. 21).

But these reforms frequently fail to engage with these critiques, and whilst a reform such as this positions itself as 'value-free science' and ideologically free, it is arguably quite the opposite. Historically, as we have seen, neuroscience has been leveraged in damaging ways. Neuroreasoning was evident during the Stolen Generation in Australia, and during the Nazi regime in Germany, as a justification for widespread sterilisation (Zeidman, 2020). Indeed neuroscience and biologism may 'generate effective oppressive technologies' (Rose & Rose, 1976, p. 122), such as types

of surveillance and authoritarianism in education (OECD, 2007), tenets which can clearly be seen within these reforms, albeit represented as ‘accreditation processes’.

In asking, where are the silences (Bacchi, 2009), it is clear that the reforms undermine the expertise of initial teacher educators, but so too, in contradictory fashion, the reforms fail to engage with critiques and limitations of the brain-based approach. The ‘Strong Beginnings’ report does not issue any limitations, caveats or warnings in regard to neuroscience or brain science, even though it references the OECD (2002) report, ‘Understanding the Brain’, which articulates numerous limitations throughout, as does the follow-up report (OECD, 2007). For instance, as the OECD explains, an overly scientific approach to education might be *questioned*: whilst on the one hand neurosciences can ‘inform education’, on the other it ‘runs the risk of creating an education system which is excessively scientific and highly conformist’ (OECD, 2002, p. 17). In the follow-up report, it advises:

Neurosciences can deliver responses in the search of a better quality in education, open to the largest number of people possible, *but* they can also generate abuse. Bruer, who was one of the first to advocate education based on neurosciences (“brain-based education”), became one of the strongest critics towards this notion (Bruer, 1993). (OECD, 2007, p. 132, our emphasis)

Thus, neuroscience can be misused, as emphasised in the OECD (2007) report on brain science, and clearly eugenics is evidence of this. However, ‘Strong Beginnings’ fails to acknowledge any risk as previously identified by the literature. This might render its claim to be based upon an ‘evidence-based’ or the ‘what works’ approach a misnomer (Malone & Hogan, 2020; Wescott, 2022). It would be reasonable for a critical reader to expect the reforms to be bracketed within limitations of this approach.

Deficit framings of students and oppressive technologies

Whilst this is not immediately evident within ‘Strong Beginnings’ (2023), we argue there is a risk that this deficit and narrow focus on brain science and binaries of ‘novice versus expert’ brains as a fixed and stable cognitive position, may be indicating how policy is making incremental steps in this direction, facilitating a movement towards pseudoscientific language, as a means to *fix* students within deficit frames (e.g. the student has ‘the brain of a novice’, as contrasted to another student who is deemed to have the ‘brain of an expert’) (Mahoney, 2023). Brain-based approaches ‘encourages [students] to develop their self-image through an external objectification, provided by neuroscientists and reinforced by educators’ (Choudhury & Wainyn, 2022, p. 44).

Gillborn (2016) argues that pseudoscientific discourses function as hidden racism, as the emphasis on genetics enables the leveraging of ‘scientific fact’ within racist discourse and deficit-setting policy-agendas. Brain-based approaches can be particularly dehumanising for students from disadvantaged backgrounds and posit risky framings, as ‘the mobilization of pseudoscientific discourses ... dehumanize the poor and promote biological theories of inferiority’ (Gillborn, 2016, p. 366). So

too, Rose and Rose (1976) argue that when all human experiences are 'subsumed into categories of stimulus and response, reward and punishment', individuals' success or failure in a competitive society is regarded as 'a result of a flaw in their chemistry or brain structure' (Rose & Rose, 1976, p. 121). The following section examines how the reforms have procured legitimacy.

How/where has this representation of the 'problem' been produced, disseminated and defended?

In asking, 'how/where has this representation of the "problem" been produced, disseminated and defended' (Bacchi, 2009), in this section we explore how the reforms have acquired authority. In alignment with critical policy sociology, we argue that in order to analyse 'Strong Beginnings', it is essential to contextualise the reforms by locating how they are mobilised within a network of power (Ozga, 2021) and identifying organisational-level actors that may hold invested interests (Ball & Junemann, 2012; Rowe and Skourdoumbis, 2019). Without this mapping and tracing of actors, an analysis is decontextualised from the power structures in which it ascertains its voice and authority. These reforms represent a struggle over knowledge within initial teacher education. Therefore, in this section we seek to illuminate how the reforms will bolster the power, role and authority of particularly intermediary organisations.

A key influencer within the 'Strong Beginnings' (2023) report is the Australian Education Research Organisation (AERO). This is significant, given that the relatively recent initiative of AERO means that their influence within major reports has been limited to non-existent. However, within 'Strong Beginnings', AERO's influence was demonstrated on both the individual level and organisational level. Their Chief Executive Officer (Jenny Donovan)³ was included within the authorship panel, and at the organisational level, AERO was commissioned to 'review and synthesise ... evidence-based practices in education' (Australian Government, 2023, p. 27) and to rate these practices against its 'standards of evidence' (for a critique of AERO, see Rowe, 2022a). Mockler (2023) points to this involvement, writing that the panel designed the 'core content' for ITE programs based upon an 'unpublished research synthesis prepared for the review' by AERO (p. 128).

In elaborating on their involvement, the report writes that AERO was engaged to.

... identify the evidence-based practices which have the greatest impact on student learning that ITE students should learn and be able to demonstrate, to inform amendments to the Accreditation Standards Education Research and Procedures (**Reform Area 1**). (Australian Government, 2023, pp. 94, authors' emphasis)

We interpreted the report as AERO being principally influential in advocating for 'reform area 1' (see above). Notably, our careful reading of the 'Strong Beginnings'

³ Note that Jenny Donovan is the former Executive Director of the *Centre for Education Statistics and Evaluation* (CESE) (see Lewis and Hogan 2019 for a discussion of CESE).

report suggests that there were disagreements between panel members, with the AERO board pushing for punitive-accountability measures:

The AERO Board highlighted a preference for *authoritative national elaborations* of core content. It argued that anything less would be *insufficient* to underpin consistent national understanding of these practices by accreditation panels and ITE program developers... [However] The Panel does not believe that a full review of the Accreditation Standards and Procedures is required to implement the core content. (Australian Government, 2023, pp. 31, authors' emphasis)

The AERO board were arguing for more 'authoritative', or punitive policy positions such as loss of accreditation or punitive funding (as demonstrated in reform area 2), in order to enforce or compel educators to carry out the proposed core content. The disempowering of initial teacher educators, and seeking to replace expert knowledge, is evident within these reforms. Clearly this is leveraged within a network of significant policy players, actors and edu-preneurs (Rowe, 2022b, 2023), and the following section considers how this is located within the 'what works' movement. This is notable, given that the 'what works' movement arguably has achieved limited influence in education policy reform in Australia until relatively recently.

AERO's advocacy for 'evidence-based' approaches and 'what works'

Shortly before the 'Strong Beginnings' published report, AERO commissioned a desktop review of 'what works' in education, which was later made available on the AERO website (Abdo et al., 2021). AERO did not actually carry out the review themselves, rather they commissioned The Centre for Evidence and Implementation (CEI) to do a 'desktop review', on the basis of 'limited Australian evidence' on the 'uptake of evidence-informed practices in education' (Abdo et al., 2021, p. 41). CEI describe themselves as 'a global, not-for-profit evidence intermediary' (p. 2). The report 'engaged with experts', with the majority drawn from their own organisations, and the remaining from 'what works' institutions (Abdo et al., 2021, p. 15). The desktop review focuses on papers published in two journals (*Implementation Science* and *Implementation Science Communications*).

It is important to note that AERO firmly positions itself as part of the 'what works' movement (Abdo et al., 2021, p. 5) working with, and alongside, 'evidence brokers' from the United States and England, with their 'evidence-based practice' pointing to the English Education Endowment Foundation (EEF) as exemplary (Abdo et al., 2021). AERO maintain strong ties with the Education Endowment Foundation as reflected in their board and publications, and the core content/standardised curriculum includes an EEF text (see Australian Government, 2023, p. 96). AERO holds up the English Endowment Foundation (EEF) as model practice (e.g. Abdo et al., 2021), but their promotion of EEF is to be expected, given AERO has ongoing ties to both the public and private-backed evidence broker (see Rowe, 2022a). AERO emerged only shortly after the commencement of EEF's five-year

'Building a global evidence ecosystem in teaching' (2018) project, led by the EEF and funded by the BHP Foundation (from the BHP Group), with the aim to embed more standardised tests in schools, and create 'EEF-style organisations' in partner countries, to drive knowledge production (see, EEF, 2024). EEF were involved in funding earlier initiatives of the 'what works' movement in Australia, having funded 'Evidence for Learning' (a subsidiary of Social Ventures Australia). It was recommended in earlier Productivity Commission Reports that AERO be modelled on the EEF (see Australian Government Productivity Commission, 2016, p. 20). So, too, the expert board of AERO continues to reflect this relationship: the founder of EEF (Sir Kevan Collins) sits on AERO's 'expert board' (AERO, 2022). These highly influential policy actors reflect the ongoing blurring of private and public (Innes, 2023). Thus, we would argue that the 'what works' movement, whilst presenting itself as 'value free', is ideological in shape and intent and driven by vested interests. As is common in England and the United States, these foundations, think tanks and intermediaries are mutually driven by private interests and private funding, and are invested in 'knowledge production' (Innes, 2023; Lubienski et al., 2011; Scott & Jabbar, 2014).

Therefore, a means to disrupt or question these representations, is to question the evidence foundation that it is reliant upon, but also the vested interests that impel the agenda. We argue, in alignment with other scholars such as Choudhury and Wannyn (2022), that many advocates or lobbyists for brain-based informed education lack any 'formal training in cognitive neuroscience' (p. 35), and rather tend to represent a variety of agendas and business interests; regarding this approach as profitable and 'profit-making by marketing [brain science] as a turnkey solution' to the various problems facing the educational system (Choudhury & Wannyn, 2022, p. 35).

Further, it is clear that these reforms maintain a strong emphasis on increasing accreditation and certain *types* of evidence. In the following section we focus on how these reforms bolster the scope, powers and strengths of the Australian Institute for Teaching and School Leadership (AITSL), and mandate core content and 'research' that has been paid for by conservative think tank (Centre for Independent Studies or CIS).

How the reforms strengthen the scope and powers of intermediary organisations

The reforms into initial teacher education have continually strengthened the scope and powers of intermediary organisations, evidence brokers and organisations with private–public partnerships, inadvertently undermining the expertise of educators and creating profitable opportunities. For example, the previously implemented LANTITE tests are a significant revenue generator for ACER.⁴ This influence of intermediaries and evidence-brokers has been growing in Australia since the 2010s, as Barnes and Cross (2021) point out. AITSL were tasked with developing the *Australian Professional Standards for Teachers* in 2011, which was notable for

⁴ LANTITE (Literacy and Numeracy Test for Initial Teacher Education students) is a significant revenue raiser for ACER, estimated as \$3.7 million a year annually (Barnes & Cross, 2020; Knott, 2016).

how it enabled AITSL to ‘become a key player in overhauling initial teacher education program(s)’ (Barnes & Cross, 2021, p. 460). So too, Savage (2016) points to ‘unprecedented policy development roles’ (p. 2) of diverse organisations and actors, such as AITSL, and potentially ‘creating new power imbalances’ (p. 15).

There are numerous examples throughout the report which highlight how AITSL’s role will be strengthened, such as recommendation one which specifies that AITSL will be tasked with adding the ‘core content’ to the Accreditation Standards and Procedures (Recommendation 1):

The Australian Institute for Teaching and School Leadership to add ‘core content’ as a schedule to the Accreditation Standards and Procedures. The schedule should *describe what initial teacher education programs should teach and the learning outcomes that should be achieved*. (Australian Government, 2023, p. 17, our emphasis)

In challenging the authority and agency of ITE educators to determine what is taught and the learning outcomes, these reforms reflect the interventionist role of AITSL, which will be coupled with increased ‘accreditation’, or to use another word - *authorisation*. The power has fundamentally shifted towards these intermediary bodies. Overall, the report will increase accreditation and accountability measures, as coupled with punitive approaches, bolstering the powers of intermediary organisations, such as AITSL, but also adding new accountability measures, such as an ‘ITE quality assurance board’ (recommendation 3) (Australian Government, 2023, p. 10).

The incremental creep of influence and intervention highlights advancing surveillance and performative accountability (Daliri-Ngametua et al., 2022), and ‘scripted forms of pedagogy’ (Hickey et al., 2022, p. 294) but so too the augmentation of powers of externalised assessors, in that AITSL will be responsible for developing and mandating a standardised curriculum, while another board will ensure the standardised curriculum is adhered to (noting that it is coupled with performance-based funding, reform area 2).

There is a risk in that the core content overly standardises ITE, even though the report recommends ‘responsive teaching’, and the two would seem incompatible. For instance, mandating and standardising core content, with prescribed core texts, scholars, and dominant ideologies, would seem significantly contradictory to the notion of ‘responsive teaching in ways that are culturally and contextually appropriate’ (Australian Government, 2023, p. 9). Whilst on one hand, the report advocates for ITE students to learn mandated, standardised curriculum which has been developed by an externalised provider (and *regulated* by a standardised provider), whereas on the other, requires that students learn about ‘responsive teaching’; the two aspects would seem to be intellectually incongruent.

Promoting ideas and paying for ideas: the Centre for Independent Studies

Think tanks demonstrate ‘the rapid infusion of private interests and influence upon public policymaking’ (Lubienski et al., 2016, p. 56) and there is demonstrably rising

influence by think tanks in education policy in Australia. The think tank, the Centre for Independent Studies (CIS), which does not make its financials public, retains an evident presence within 'Strong Beginnings' (2023). CIS are a 'conservative right wing advocacy think tank' who actively produce reports that are 'dressed up as research' (Lingard, 2015, p. 15), and in this respect are different to US-style think tanks who may only seek to promote ideas that are put forward, rather than *produce* ideas (Lubienski et al., 2016).

The 'Strong Beginnings' reforms, in particular the 'brain and learning', establishes mandated curriculum with prescribed texts and references. Educational psychologist and university professor John Sweller's work comprises a major part of the mandated references (Australian Government, 2023, p. 96). Sweller is well-known for 'cognitive load theory', and his work is promoted by the English Education Endowment Foundation (EEF, 2023) and the Centre for Independent Studies (CIS, 2023). The shared agenda is evident within the prescribed texts, and CIS-commissioned research is mandated as a 'key reference' within the 'core content' (Australian Government, 2023, p. 96). This is concerning, as initial teacher educators will be mandated to include publications as developed, and paid for, by a conservative think tank.

There are similarities here with the United States context, as several intermediaries, evidence brokers and think tanks work in alignment around 'idea orchestration' (Lubienski et al., 2016), in advancing particular ideas. For instance, Dr Jennifer Buckingham has widely promoted 'Strong Beginnings' (2023) and its associated reforms (see Buckingham, 2023). Buckingham is a prominent researcher with high-impact, with long-held ties to the Centre for Independent Studies as a senior researcher and board member, amongst other roles (CIS, 2024). She also occupies the non-executive Director position at AITSL (AITSL, 2017). In this way, there are multiple ties advancing a shared ideological agenda, demonstrated at the national and global scale.

Concluding discussion

In this paper, we analysed the 'Strong Beginnings' report (Australian Government, 2023), which is significant in introducing major reforms into initial teacher education. This reform will embed standardised curriculum into initial teacher education, and tie this curriculum to graduate teacher standards, enforcing an emphasis on neuroscience, 'the brain and learning' as core content. Whilst the report was initiated in the context of a major teaching shortage in Australia, it fails to engage with the most cited reason for teachers leaving the profession (workload) (Creagh et al. 2023; Fitzgerald et al., 2018; Heffernan et al., 2022; Stacey et al., 2023; Stacey et al., 2020). Rather, it pushes a 'what works' agenda, advocating for increased accreditation, 'evidence' and neuroscience reasoning. Students are to learn about mental models and schemas, the features of biologically primary versus biologically secondary knowledge acquisition, and novice versus expert learners (Australian Government, 2023). This is significant, given that it is the first time in policy reform in

Australia in which neuroscience and a brain-based approach to learning has been mandated into initial teacher education at a nation-wide level.

In pointing to the limitations of a brain-based approach to learning, and historical underpinnings of eugenics, in this paper we have argued there is risk of these reforms solidifying hidden racisms, and deficit fixed tropes of low socio-economic status students, potentially through the use of artificial intelligence. Within ‘Strong Beginnings’, the push for the brain-based approach is not bracketed within or coupled with any limitations that have been previously discussed (e.g. OECD, 2002, 2007), despite the history of neuroscience within eugenics.

We point to how the reforms are influenced by networked intermediary organisations, including think tanks and ‘evidence brokers’. Certainly, the push for a neuroscience approach was authored and primarily influenced by the Australian Education Research Organisation (AERO). Herein, the link to a network of private/public actors is evident. The core content is linked to prescribed references, which predominantly includes texts by John Sweller, whose work is promoted by the conservative think tank Centre for Independent Studies (CIS), and the English Education Endowment Foundation (EEF). The standardised curriculum will include ‘research’ that has been commissioned by the Centre for Independent Studies (Sweller, 2021), that initial teacher educators must include in their curriculum (Australian Government, 2023, p. 96). Clearly, the reforms will bolster the scope, powers and strengths of evidence brokers, intermediary organisations and private interests, including the Australian Institute for Teaching and School Leadership (AITSL), the Australian Education Research Organisation (AERO)⁵ and Centre for Independent Studies (CIS), and indicate ongoing influence by the English Education Endowment Foundation (EEF).

The danger in this shift in emphasis from viewing ‘education as a human event of communication, meaning making and interpretation towards *cause-effect relationships*’ (Biesta et al., 2015, p. 11) is of profound relevance for all involved in teacher education in Australia. It is also important for those working in the field globally where there is an active destruction and overthrow of the established ontology, axiology and praxeology (Biesta et al., 2015) of a university based ITE, particularly evident in England and the embedding of a school-led system of ITE (Mutton et al., 2017). In the effort to make the preparation of teachers more reliant on school-based clinical experience/s coupled with the ‘what works’ movement, and certainties of neuroscience, the nature of teaching and learning and, what it means to be a teacher, is fundamentally undermined (Mockler & Stacey, 2021). The standardised curriculum not only disempowers initial teacher educators, but so too it discounts other forms of teacher knowledge (Larsen & Mockler, 2023; McKnight & Morgan, 2019; Mockler & Stacey, 2021). Wescott (2022) refers to this as a ‘burden of proof’ that is put onto teachers, discounting other forms of proof they may collect in the course of their teaching day.

Finally, we are not arguing that science is an ineffective investigative tool; but the application of rigid positivistic principles and a mechanistic worldview cannot account for the multifaceted human actions that form the human experience. This

⁵ Note that AITSL, ACARA and AERO are each underpinned by the registered charity, Educational Services Australia (ESA).

means acknowledging that the sociology of schools and of classrooms necessitates a broad-based epistemological foundation reflecting an attention to detail and a criticality within the 'evidence era', of how evidence is constructed (Helgetun & Menter, 2022).

Author contributions Both A/P Skourdoumbis and Dr. Rowe have contributed equally to this article.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions. N/A. No funding was received for this article.

Data Availability Any of the datasets (eg: The Strong Beginnings Report) are publicly available and can be found through the following link: <https://www.education.gov.au/quality-initial-teacher-education-review/resources/strong-beginnings-report-teacher-education-expert-panel>

Declarations

Competing interests There are no interests around the production of this article of a financial or personal nature.

Ethical Approval N/A.

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Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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